

ASSIGNMENT

DATA ANALYTICS PROCESS

AND INTERPRETATION

IS4116 Business Intelligence Systems



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Github link: https://github.com/Rashi990/IS4116_BIS_Assignment.git

1. Business Domain & Dataset Selection

- Business Domain : E-Commerce
- Dataset : E-commerce Customer Behavior Dataset
- Dataset link : <https://www.kaggle.com/datasets/uom190346a/e-commerce-customer-behavior-dataset>

1.1 Business Domain

The selected business domain is E-Commerce, specifically focusing on Customer Behavior Analytics in an e-commerce company. The goal is to analyze customer data to understand purchasing patterns, customer segmentation, spending behavior and factors influencing customer satisfaction to develop targeted marketing strategies.

1.2 Dataset Selection

I have selected above “E-commerce Customer Behavior Dataset” from Kaggle and it provides a comprehensive view of customer behavior within an e-commerce platform. The dataset consists of key customer attributes including Customer ID, Gender, Age, City, Membership Type, Total Spend, Items Purchased, Average Rating, Discount Applied, Days Since Last Purchase, Satisfaction Level. These variables help in identifying different customer segments based on spending patterns and shopping behavior.

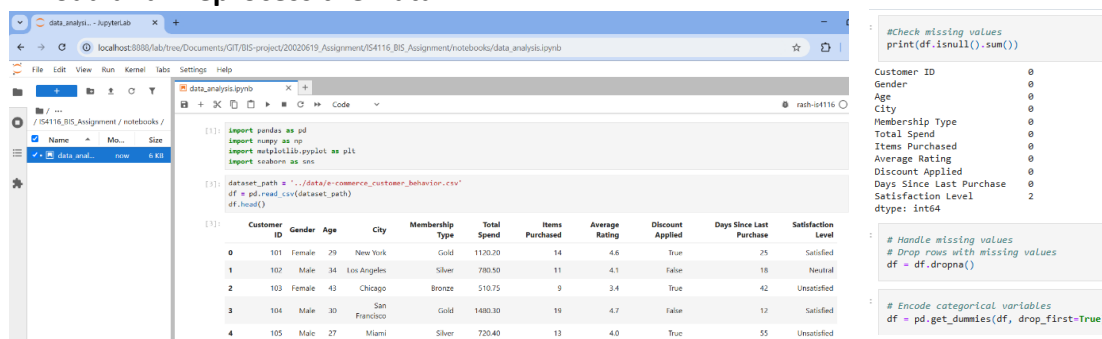
2. Analytical Scenario & Process

2.1 Business Question

The primary business question is: **How can we segment customers based on their purchasing behavior to create targeted marketing strategies?** The company wants to segment its customers to optimize marketing strategies improving their engagement and retention

2.2 Data Analytics Process Using Python

Step 1: Load and Preprocess the Data



```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

[2]: dataset_path = "../data/e-commerce_customer_behavior.csv"
df = pd.read_csv(dataset_path)
df.head()
```

	Customer ID	Gender	Age	City	Membership Type	Total Spend	Items Purchased	Average Rating	Discount Applied	Days Since Last Purchase	Satisfaction Level
0	101	Female	29	New York	Gold	1120.20	14	4.6	True	25	Satisfied
1	102	Male	34	Los Angeles	Silver	780.50	11	4.1	False	18	Neutral
2	103	Female	43	Chicago	Bronze	510.75	9	3.4	True	42	Unsatisfied
3	104	Male	30	San Francisco	Gold	1480.30	19	4.7	False	12	Satisfied
4	105	Male	27	Miami	Silver	720.40	13	4.0	True	55	Unsatisfied

```
# Check missing values
print(df.isnull().sum())

Customer ID      0
Gender           0
Age             0
City            0
Membership Type  0
Total Spend      0
Items Purchased  0
Average Rating   0
Discount Applied 0
Days Since Last Purchase 0
Satisfaction Level 2
dtype: int64

# Handle missing values
# Drop rows with missing values
df = df.dropna()

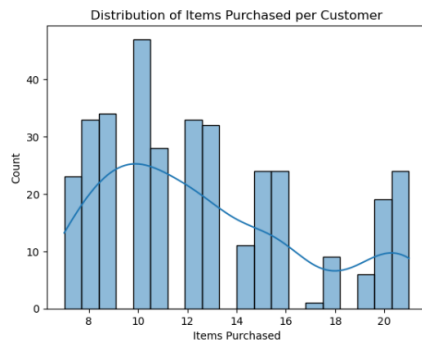
# Encode categorical variables
df = pd.get_dummies(df, drop_first=True)
```

First loaded the dataset from csv file to Jupyter notebook in Anaconda environment. Then missing values were checked and handled appropriately, categorical variables were encoded.

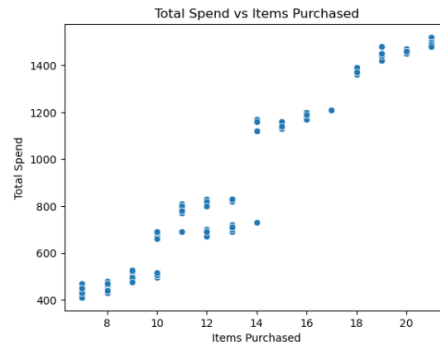
Step 2: Exploratory Data Analysis (EDA) & Visualizations

- Analysis of the distribution of purchasing with total spend and customers.

```
# Items Purchased Distribution
sns.histplot(df['Items Purchased'], bins=20, kde=True)
plt.title('Distribution of Items Purchased per Customer')
plt.show()
```



```
# Scatter plot for Total Spend vs Items Purchased
sns.scatterplot(x=df['Items Purchased'], y=df['Total Spend'])
plt.title('Total Spend vs Items Purchased')
plt.xlabel('Items Purchased')
plt.ylabel('Total Spend')
plt.show()
```

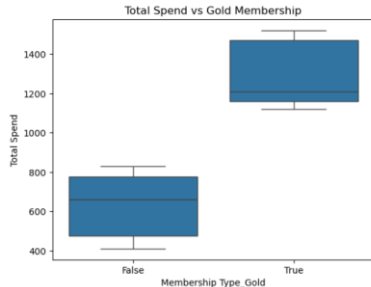


```
# Line plot for Total Spend vs Days Since Last Purchase
sns.lineplot(x=df['Days Since Last Purchase'], y=df['Total Spend'])
plt.title('Total Spend vs Days Since Last Purchase')
plt.xlabel('Days Since Last Purchase')
plt.ylabel('Total Spend')
plt.show()
```

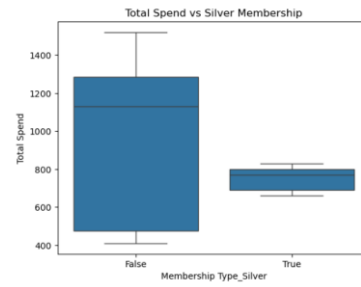


- Analysis of the distribution of membership types with total spend.

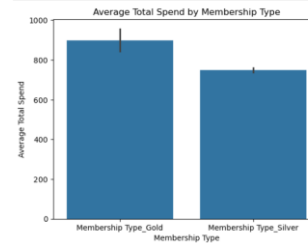
```
# Boxplot for Membership Type_Gold vs Total Spend
sns.boxplot(x='Membership Type_Gold', y='Total Spend', data=df)
plt.title('Total Spend vs Gold Membership')
plt.show()
```



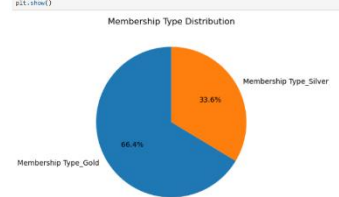
```
# Boxplot for Membership Type_Silver vs Total Spend
sns.boxplot(x='Membership Type_Silver', y='Total Spend', data=df)
plt.title('Total Spend vs Silver Membership')
plt.show()
```



```
# Bar plot for Membership Type vs Average Total Spend
sns.barplot(x=df['Membership Type_Gold'], y=df['Average Total Spend'], data=df)
plt.title('Average Total Spend by Membership Type')
plt.xlabel('Membership Type')
plt.ylabel('Average Total Spend')
plt.show()
```

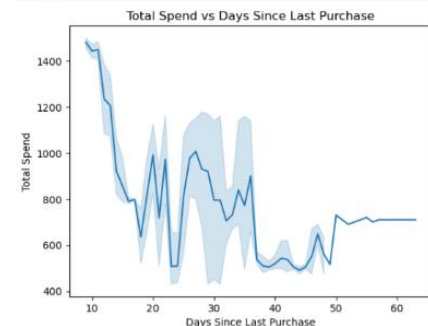


```
# Pie chart for Membership Type distribution
membership_counts = df['Membership Type_Gold'].value_counts()
membership_counts.index = ['Membership Type_Gold', 'Membership Type_Silver']
plt.pie(membership_counts, labels=membership_counts.index, autopct='%1.1f%%', startangle=90)
plt.show()
```

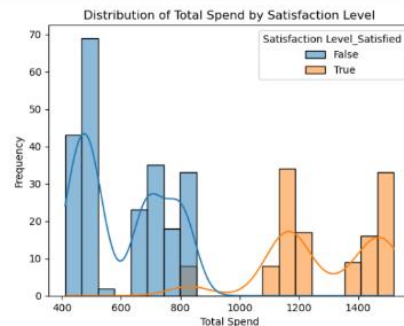


- Analysis of Total spend vs Days Since Last purchase

```
# Line plot for Total Spend vs Days Since Last Purchase
sns.lineplot(x=df['Days Since Last Purchase'], y=df['Total Spend'])
plt.title('Total Spend vs Days Since Last Purchase')
plt.xlabel('Days Since Last Purchase')
plt.ylabel('Total Spend')
plt.show()
```

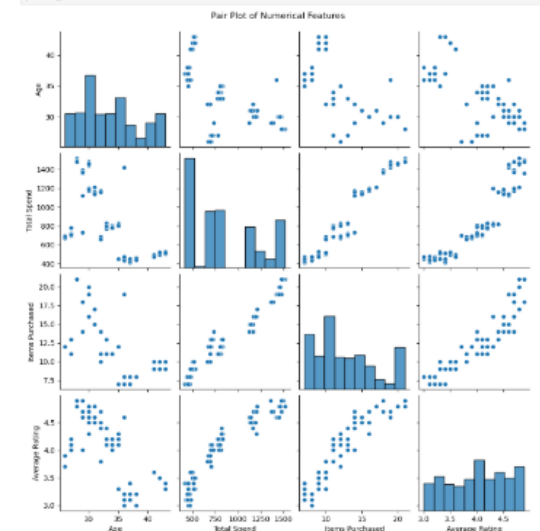


```
# Histogram of Total Spend for Satisfied vs Unsatisfied Customers
sns.histplot(df, x='Total Spend', hue='Satisfaction Level', kde=True, bins=20)
plt.title('Distribution of Total Spend by Satisfaction Level')
plt.xlabel('Total Spend')
plt.ylabel('Frequency')
plt.show()
```



- Analysis of Total spend by Satisfaction levels

```
# Pair plot for numerical features
sns.pairplot(df[['Age', 'Total Spend', 'Items Purchased', 'Average Rating']])
plt.show()
```

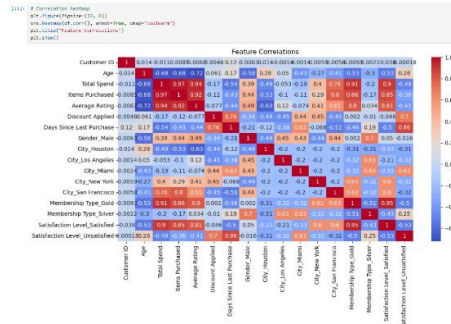
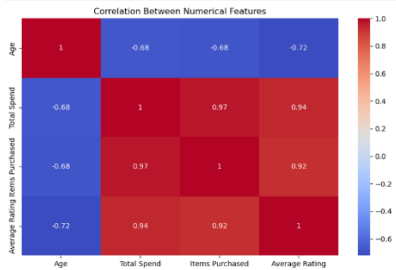


- Analysis of Numeric features

Step 3: Statistical Analysis & Visualizations

- Correlation heatmaps of Numerical Features

```
# Correlation heatmap
plt.figure(figsize=(10, 8))
sns.heatmap(df[['Age', 'Total Spend', 'Items Purchased', 'Average Rating']], annot=True, cmap='coolwarm')
plt.title('Correlation Between Numerical Features')
plt.show()
```



```
# Linear Regression
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Define features and target
X = df[['Age', 'Items Purchased', 'Average Rating', 'Discount Applied', 'Days Since Last Purchase']] # Features
y = df['Total Spend'] # Target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train model
model = LinearRegression()
model.fit(X_train, y_train)

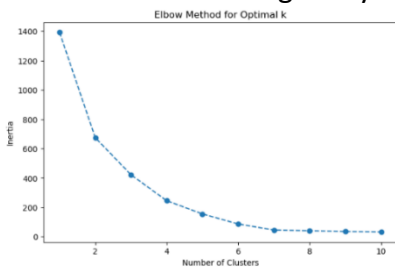
# Predictions
y_pred = model.predict(X_test)

# Evaluation
print("R-squared:", r2_score(y_test, y_pred))
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))

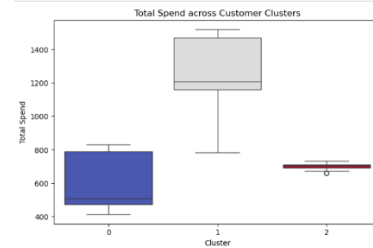
R-squared: 0.9853119321779633
Mean Squared Error: 2104.240356783039
```

- Regression analysis

- Clustering Analysis



```
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='Cluster', y='Total Spend', hue='Cluster', palette='coolwarm', legend=False)
plt.title('Total Spend across Customer Clusters')
plt.show()
```



- By applying K-Means, customers were segmented into 3 groups based on their spending and purchasing behavior. Targeted marketing strategies can be developed for each segment.

Cluster Interpretation

Cluster	Characteristics	Business Strategy
Cluster 0	Low spending, few purchases	Offer promotions, encourage repeat purchases
Cluster 1	Medium spending, average purchases	Provide loyalty programs, personalized discounts
Cluster 2	High spending, frequent purchases	Premium membership, exclusive deals

3. Interpretation & Business Implications

- High-value customers drive revenue** – Focus on retaining Cluster 2 customers with premium services, exclusive deals, and VIP benefits.
- Total Spend & Items Purchased are key indicators** – Increase the number of items per purchase using bundles, bulk discounts, and personalized product recommendations.
- Low-spending customers need engagement** – Use promotions, loyalty programs, and referral incentives to encourage repeat purchases and boost spending.
- Medium spenders offer upselling opportunities** – Provide exclusive deals, subscription-based services, and targeted promotions to move them toward higher spending.
- Personalization is crucial for customer retention** – Implement AI-driven recommendations, targeted ads, and personalized communication to enhance engagement and loyalty.